Summary in Graph

Exam Summary (GO Classes GATE CS/DA | DBMS | Relational Algebra | Weekly Quiz 10)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (51) out of 513 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Databases

Consider the join of a relation R with a relation S. If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

A. $m+n$ and 0
B. mn and 0
C. $m+n$ and $|m-n|$

D. mn and $m+n$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

The relation schema, Person (pid, city), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following two relations, $R(A, B)$ and $S(A, C)$:

R	
A	B
10	20
20	30
30	40
30	50
50	95

S	
A	C
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$\sigma_{B<C}\left(R \bowtie_{R.A=S.A} S\right)$ is _____.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Given two union compatible relations $R_1(A, B)$ and $R_2(C, D)$, what is the result of the operation $R_1 \bowtie_{A=C \wedge B=D} R_2$?

A. $R_1 \cup R_2$

B. $R_1 \times R_2$

C. $R_1 - R_2$

D. $R_1 \cap R_2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where $A1, A2$ are sets of attributes in r with $A1 \subset A2$ and $F1, F2$ are Boolean expressions based on the attributes in r?

A. $\pi_{A1}(\sigma_{(F1 \wedge F2)}(r))$
B. $\pi_{A1}(\sigma_{(F1 \vee F2)}(r))$
C. $\pi_{A2}(\sigma_{(F1 \wedge F2)}(r))$
D. $\pi_{A2}(\sigma_{(F1 \vee F2)}(r))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following relations $P(X, Y, Z), Q(X, Y, T)$ and $R(Y, V)$.

Table: P

X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Table: Q

X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

Table: R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?
$\Pi_x(\sigma_{(P.Y=R.Y \wedge R.V=V2)})(P \times R)) - \Pi_x(\sigma_{(Q.Y=R.Y \wedge Q.T>2)})(Q \times R))$ Answer: _____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The following relation records the age of 500 employees of a company, where $empNo$ (indicating the employee number) is the key:

$$empAge(\underline{empNo}, age)$$

Consider the following relational algebra expression:

$$\Pi_{empNo}(empAge \bowtie_{(age>age1)} \rho_{empNo1,age1}(empAge))$$

What does the above expression generate?

A. Employee numbers of only those employees whose age is the maximum
B. Employee numbers of only those employees whose age is more than the age of exactly one other employee
C. Employee numbers of all employees whose age is not the minimum
D. Employee numbers of all employees whose age is the minimum

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a database that includes the following relations:

Defender(name, rating, side, goals)
Forward(name, rating, assists, goals)
Team(name, club, price)

Which ONE of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where ϕ denotes the empty set?

A. $\Pi_{\text{name}}(\text{ Team })\setminus(\Pi_{\text{name}}(\text{ Defender })\cap\Pi_{\text{name}}(\text{ Forward }))=\phi$
B. $(\Pi_{\text{name}}(\text{ Defender })\cap\Pi_{\text{name}}(\text{ Forward }))\setminus\Pi_{\text{name}}(\text{ Team })=\phi$
C. $\Pi_{\text{name}}(\text{ Team })\setminus(\Pi_{\text{name}}(\text{ Defender })\cup\Pi_{\text{name}}(\text{ Forward }))=\phi$
D. $(\Pi_{\text{name}}(\text{ Defender })\cup\Pi_{\text{name}}(\text{ Forward }))\setminus\Pi_{\text{name}}(\text{ Team })=\phi$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

In a relational database there are three relations:

- $Customers = C$(CName),
- $Shops = S$(SName),
- $Buys = B$(CName, SName).

Which of the following relational algebra expressions returns the names of shops that have no customers at all? [Here Π is the projection operator.]

A. $\Pi_{\text{SName}}B$
B. $S - B$
C. $S - \Pi_{\text{SName}}B$
D. $S - \Pi_{\text{SName}}((C \times S) - B)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

In a relational database there are three relations:

- $Customers = C$(CName)
- $Shops = S$(SName)
- $Buys = B$(CName, SName)

Then the Relational Algebra expression (Π is the projection operator).

$$C - \Pi_{\text{CName}}((C \times S) - B)$$

returns the names of

A. Customers who buy from at least one shop.
B. Customers who buy from at least two shops.
C. Customers who buy from all shops.

D. Customers who do not buy buy anything at all.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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